Cancelable Biometrics Realization With Multispace Random Projections

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***Abstract*—Biometric characteristics cannot be changed; therefore, the loss of privacy is permanent if they are ever compromised. This paper presents a two-factor cancelable formulation, where the biometric data are distorted in a revocable but nonreversible manner by first transforming the raw biometric data into a fixed-length feature vector and then projecting the feature vector onto a sequence of random subspaces that were derived from a user-specific pseudorandom number (PRN). This process is revocable and makes replacing biometrics as easy as replacing PRNs. The formulation has been verified under a number of scenarios (normal, stolen PRN, and compromised biometrics scenarios) using 2400 Facial Recognition Technology face images. The diversity property is also examined.**

***Index Terms*—Cancelable biometrics, face biometrics, privacy, random projection (RP).**

I. Introduction

IF BIOMETRICS is compromised, it is compromised forever because it cannot be changed. Cancelable biometrics has been proposed, where the biometric image is distorted in a repeatable but nonreversible manner before template generation. If the cancelable template is compromised, the distortion characteristics are changed, and the same biometrics is mapped to a new template, which is used subsequently. To formally define the concept, four principal objectives were outlined here [1].

1) Diversity: No same cancelable template can be used in two different applications.
2) Reusability: Straightforward revocation and reissue in the event of compromise.
3) One-way transformation: Noninvertibility of template computation to prevent recovery of biometric data.
4) Performance: The formulation should not deteriorate the recognition performance.

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A. Related Research

Davida *et al.* [2] made the first attempt toward this direction, and Ratha *et al.* [3] concretized the idea of cancelable biometrics. Davida *et al.* proposed a majority decoding scheme for iris biometrics. In their scheme, a pair of related binary representations of the iris code, i.e., the input and test template, which is 2048 bits in length, was extracted through the majority decoding scheme and matched by using hamming distance. Subsequently, algebraic decoding was applied in order to rectify the offset of the test data via an offline token. To a certain extent, the scheme may preserve user privacy, as the biometric template was noninvertible. However, neither the issues of reusability nor practical work has been addressed in this scheme. Juels [4], [5] generalized and improved the scheme of Davida *et al.* through a modification in error-correcting codes and hence reduced the code size and achieved higher resilience. Clancy *et al.* [6] implemented the technique that was proposed by Juels *et al.* In Clancy *et al.*'s work, a group of minutiae points was extracted from an input fingerprint to bind in a locking set using a polynomial-based secret-sharing scheme. Subsequently, a nonrelated chaff point was added intentionally to "shadow" the key to maximize the unlocking computational complexity, where the secret key could only be recovered if there is a substantial overlap between the input and the testing fingerprint. The method has been theoretically proven to be secure in protecting the secrecy of the fingerprint. Nevertheless, it is way beyond the level of practical use due to the high false reject rate (FRR) at 20%–30%. Juels and Sudan's fuzzy vault scheme has spurred great interest among researchers; a few methods such as geometric hashing [7], [8] were proposed to solve the template-query alignment issue. Other variations of fuzzy vault schemes such as [9] have also been reported.

Ratha *et al.* [3] introduced an intentional distortion of a biometrics signal that is based on a chosen transform function. The biometrics signal was distorted in the same fashion at each presentation, i.e., during enrolment and for every subsequent authentication. With this approach, every instance of enrolment can use a different transform function, thus rendering cross matching impossible. Furthermore, if one variant of the biometrics is compromised, then the transformation can simply be changed to create a new variant for reenrolment. However, it is not an easy task to design such a function due to the characteristics of the feature vector. Generally, extracted features that take different values that change in some range, rather than take precise values, depend on the type of biometrics used and feature extractor; therefore, the transform function has to satisfy some smoothness criteria. While providing robustness

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against the variability of the same user's biometric data, that transformation also has to distinguish different users successfully. Sutcu *et al.* [10] realized this idea by proposing a sum of weighted and shifted Gaussian functions as a noninvertible and scalable transformation function. However, their preliminary results showed that the method was not so favorable in terms of recognition performance. Tulyakov *et al.* [11] presented a method of hashing fingerprint minutiae information and performing fingerprint matching in a new domain. Since only hashed data are transmitted and stored in the server database, it is impossible to restore fingerprint minutiae locations using hashed data. Ang *et al.* [12] proposed a similar technique with key-dependent transformation, so that the matching can be done in the transformed domain. Yet, both transforms degrade the matching accuracy significantly in the altered domain.

Linnartz and Tuyls [13] assume that a noise-free template X of a biometric identifier is available at the enrolment time and use this to enroll a secret S to generate a helper data W. They assume that each dimension of the template is quantized at q resolution levels. In each dimension, the process of obtaining W is equivalent to finding residuals that must be added to X to fit to odd or even grid quantum, depending on whether the corresponding S bit is zero or one. At verification time, the (noise-prone) biometric template Y is used to obtain S', which is approximately the same as S. It is hoped that the relatively few errors in S' can be corrected using error-correction techniques. The proposed technique assumes that the noise in each dimension is relatively small compared to quantization Q. Tuyls *et al.* [14] reported a practical system based on the proposed scheme; however, the secret bit length generated (40 bits) is still quite low for most security applications.

Soutar *et al.* [15] offers a different approach for generating cancelable biometrics from fingerprints using optical computing techniques. The templates in this case are correlation patterns that are derived from training images, which are subsequently mixed with random data to generate identification codes. However, the scheme did not explain in a satisfactory manner the security of the transformations, and no results were published. Goh and Ngo [16], and Teoh and Ngo [17] introduced cancelable biometrics/key via an iterative inner product between randomized tokens and biometric data, which is advantageous compared to that of Soutar *et al.*, as the step is a one-way process. However, their formulation suffered in terms of verification performance when the genuine token was stolen and used by the impostor in the claim to be the genuine user. Savvides *et al.* [18] proposed a cancelable biometrics scheme that encrypted the training images that were used to synthesize the correlation filter for biometrics authentication. They demonstrated that convolving the training images with any random convolution kernel prior to building the biometric filter does not change the resulting correlation output peak-to-sidelobe ratios, thus preserving the authentication performance. However, the security will be jeopardized via a deterministic deconvolution with a known random kernel. Most recently, Boult [19] introduced the biometrics-based tokens that support robust distance computations, which provide cryptographic security such that it can be canceled or revoked and replaced with a new one. Note that Soutar *et al.* and Savvides *et al.* applied a set of common random numbers for all users, whereas Goh *et al.* and Teoh *et al.* make use of user-specific random numbers.

B. Our Approach

We believe that multifactor authentication is one of the solutions in addressing the more demanding need for security in authentication technology today. In this paper, we follow the direction of Goh and Teoh in the introduction of refreshness into the biometrics template via a set of user-specific random numbers from biometrics data. However, the generic two-factor noninvertible mixing process is carried out by linear subspace projection, which is known as multispace random projections (MRPs). The random subspace is constituted by the independent, zero-mean, and unit-variance Gaussian-distributed random bases, which can be generated from the user-specific pseudorandom numbers (PRNs). Since different users hold their own set of PRNs, the formulation can be extended to include multiple users or applications to produce multiple random subspaces. MRP inspires a two-factor authentication mechanism by introducing a secondary authentication factor—a private token that constitutes the user PRN—in the traditional biometrics recognition system. A two-factor authentication system has the advantage of avoiding any attack with a single factor that attempts with a stolen token or fake biometrics that works with the traditional system. This is a typical attack to the traditional token-based system, where a hacker intrudes the system with a stolen or dictionary-generated password. MRP fortifies the security level of the system, where legitimate access requires a valid token and a genuine face's feature.

From the performance perspective, we consider three different scenarios when MRP is applied.

1) The genuine's biometrics is mixed with the user-specific token (*legitimate-token* scenario hereinafter).
2) Stolen-token, in which an impostor has an access genuine token, which was used by the impostor in the claim to be the genuine user.
3) Stolen-biometrics, in which an impostor possesses intercepted biometric data of sufficiently high quality to be considered genuine.

We will show that MRP satisfies the performance requirement in these three scenarios and fulfills all the other cancelable biometrics criteria, such as diversity, reusability, and noninvertible properties, either theoretically or experimentally. We utilize face biometrics as the subject of study in this paper since there is an established and standardized database for comparison.

The outline of this paper is given as follows. Section II describes the overview of the MRP formulation. Section III describes its mathematical framework and the statistical analysis. Section IV presents the experimental results and the discussion. Section V provides the security analysis of MRP. Conclusions are drawn in Section VI.

II. Overview of MRP Formulation

MRP comprises two stages: 1) feature extraction and 2) MRPs. In the feature extraction stage, the individual's

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biometric image $\boldsymbol{i}$ is transformed to a fixed-length feature vector with length n by using a feature extractor $\boldsymbol{x} \in \Re^n$. The biometric feature vector $\boldsymbol{x}$ is further projected onto a random subspace, as determined from an externally derived pseudorandom sequence $\boldsymbol{R} \in \Re^{mn}$, where $m \leq n$. The random projection (RP) vector $\boldsymbol{v}$ is described as follows:

$$\boldsymbol{v} = \sqrt{(1/m)}\, \boldsymbol{R}\boldsymbol{x} \tag{1}$$

where $\boldsymbol{R}$ is an $m \times n$ random matrix, $m \leq n$, and n is the biometrics feature length. In the literature, RP is a simple yet powerful dimension reduction technique [20] that uses random matrices $\boldsymbol{R}$ to project the raw data into low-dimensional spaces.

RP theory is addressed by means of the Johnson–Lindenstrauss (JL) lemma [21].

1) JL Lemma: For any $0 < \varepsilon < 1$ and any integer k, let m be a positive integer such that $m \geq 4 \ln k/(\varepsilon^2/2 - \varepsilon^3/3)$. Then, for any set S of $n = |S|$ data points in $\Re^n$, there is a map $f : \Re^n \to \Re^m$ such that for all $\boldsymbol{x}, \boldsymbol{y} \in S$, we have

$$(1-\varepsilon)\|\boldsymbol{x}-\boldsymbol{y}\|^2 \leq \|f(\boldsymbol{x}) - f(\boldsymbol{y})\|^2 \leq (1+\varepsilon)\|\boldsymbol{x}-\boldsymbol{y}\|^2.$$

The lemma states that any set of k points in n-dimensional *Euclidean space* can be embedded into an $O(\ln k/\varepsilon^2)$-dimensional space, such that the pairwise distance of any two points is maintained within an arbitrarily small factor. This pleasant property implies that it is possible to alter the data's original form but still preserves it statistical characteristics.

The basic formulation can be extended to include g users to generate g different RP features, i.e., multiple random subspaces $\boldsymbol{R}^k$, which render a set of unique feature vectors

$$\boldsymbol{v}^k = \sqrt{(1/m)}\, \boldsymbol{R}^k \boldsymbol{x}^k \qquad \text{where } k = 1, \ldots, g. \tag{2}$$

III. MRPs

In this section, we analyze the MRP in terms of its feature's intraclass and interclass variations. Independent of JL lemma, we will first show that the statistical properties—mean and variance—of the feature's intraclass variation are preserved exactly like that in the feature-vector level (before RP is performed) while amplifying interclass variations via projection onto uncorrelated random subspaces in legitimate-token and stolen-biometrics scenarios. On the other hand, we show that the recognition performance remained as sole biometrics performance in the stolen-token scenario, through the choice of dissimilarity measure, i.e., the normalized inner product that governs the statistics preservation transformation. For these to be done, we need to understand the statistical properties of the product of random matrices.

A. Statistical Properties of the Product of Random Matrices

Let $\boldsymbol{R}$ be an $m \times n$ $(m < n)$ random matrix such that each entry r_{ij} of $\boldsymbol{R}$ is independent and identically distributed (i.i.d.) drawn from a Gaussian distribution with mean $\mathrm{E}[r_{ij}] = 0$ and variance $\mathrm{var}[r_{ij}] = 1$, and let ε_{ij} be the entry of $\boldsymbol{R}^T\boldsymbol{R}$. Then, $\varepsilon_{ij} = \sum_{k=1}^m r_{ki} r_{kj}$, and hence, its expectation [22] is

$$\mathrm{E}[\varepsilon_{ij}] = \mathrm{E}\left[\sum_{k=1}^m r_{ki} r_{kj}\right] = \sum_{k=1}^m \mathrm{E}[r_{ki} r_{kj}]$$

$$= \begin{cases} \sum\limits_{k=1}^m \mathrm{E}[r_{ki}]\mathrm{E}[r_{kj}], & \text{if } i \neq j \\ \sum\limits_{k=1}^m \mathrm{E}\left[r_{ki}^2\right], & \text{if } i = j. \end{cases}$$

Note that $\mathrm{E}(r_{ij}) = 0$ and $\mathrm{E}(r_{ij}^2) = 1$; hence

$$\mathrm{E}[\varepsilon_{ij}] = \begin{cases} 0, & \text{if } i \neq j \\ m, & \text{if } i = j \end{cases} \text{ or } \mathrm{E}[\boldsymbol{R}^T\boldsymbol{R}] = m\boldsymbol{I} \tag{3}$$

or if each column vector of $\boldsymbol{R}$ is normalized to have a unit length, we have

$$E[\varepsilon_{ij}] = \begin{cases} 0, & \text{if } i \neq j \\ 1, & \text{if } i = j. \end{cases} \tag{4}$$

The preceding result suggests that column vectors of $\boldsymbol{R}$ are orthogonal to each other, and this is specifically true in a high-dimensional space. This observation was also verified by Hecht-Nielsen [23]: In a high-dimensional space, there exists a much larger number of almost orthogonal than orthogonal directions. Thus, vectors having random directions might be sufficiently close to orthogonal, and equivalently, $\boldsymbol{R}^T\boldsymbol{R}$ would approximate an identity matrix. Note that $\boldsymbol{R}$ has not undergone orthonormalization process such as Gram–Schmidt's procedure due to its heavy computation, which will jeopardize the MRP for practical use.

From (3), since $\mathrm{E}[\varepsilon_{ii}] = m$ and it can be proven that $\mathrm{E}[\varepsilon_{ii}^2] = (2m + m^2)$, thus, $\mathrm{var}(\varepsilon_{ii}) = 2m$ $\forall i$. Similarly, we have $\mathrm{E}(\varepsilon_{ij}^2) = m$ and $\mathrm{E}(\varepsilon_{ij}) = 0$; thus, $\mathrm{var}(\varepsilon_{ij}^2) = m$ $\forall i$, j, $i \neq j$. Collectively

$$\mathrm{var}(\varepsilon_{ij}) = \begin{cases} m, & \text{if } i \neq j \\ 2m, & \text{if } i = j. \end{cases} \tag{5}$$

For MRP formulation, an individual random subspace can be easily extended to include g users with g PRN sets to generate g random subspaces. For instance, let $\boldsymbol{R}$ and $\boldsymbol{S}$ be two $m \times n$ $(m < n)$ uncorrelated random matrices such that each entry $r_{ij} \in \boldsymbol{R}$ and $s_{ij} \in \boldsymbol{S}$ is i.i.d. drawn from the standard Gaussian distribution, and let η_{ij} be the entry of $\boldsymbol{R}^T\boldsymbol{S}$; then, $\eta_{ij} = \sum_{k=1}^m r_{ki} s_{kj}$. Thus

$$\mathrm{E}[\eta_{ij}] = \mathrm{E}\left[\sum_{k=1}^m r_{ki} s_{kj}\right] = \sum_{k=1}^m \mathrm{E}[r_{ki} s_{kj}]$$

since $r_{ij} \in \boldsymbol{R}$ and $s_{ij} \in \boldsymbol{S}$ are uncorrelated. Hence, $\mathrm{E}(r_{ij}) = 0$, and $\mathrm{E}[\eta_{ij}] = \sum_{k=1}^m \mathrm{E}[r_{ki}]\mathrm{E}[s_{kj}]$ $\forall i, j$.

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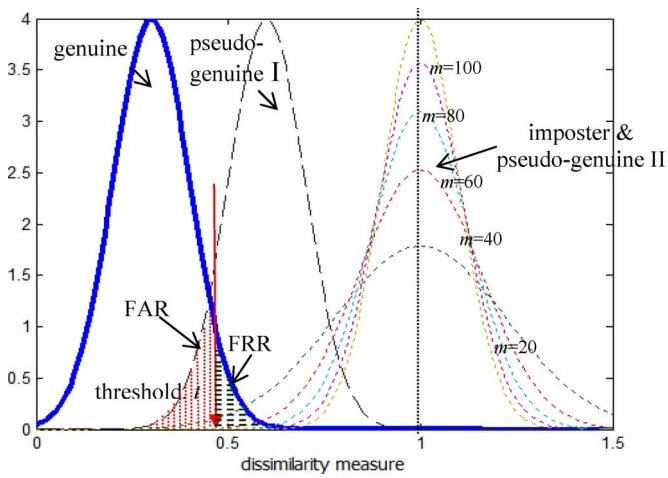

Fig. 1. Theoretical genuine, impostor, and pseudogenuine type-I and type-II distributions for MRP.

Thus, we have

$$E[\eta_{ij}] = 0 \qquad \forall i, j \text{ or } \mathrm{E}[\boldsymbol{R}^T\boldsymbol{S}] = \boldsymbol{0} \tag{6}$$

and

$$\mathrm{var}[\eta_{ij}] = m \qquad \forall\, i, j. \tag{7}$$

B. MRP's Genuine and Impostor Distribution Analysis

The nature of the biometrics system offers two possible error outcomes that are known as false accept rate (FAR) and FRR. By manipulating the decision criteria, the relative probabilities of these two outcomes can be adjusted in a way that reflected their associated cost and benefits. The interdependence of FAR and FRR is regulated by the system threshold value t, as indicated in Fig. 1. As t moves to the right, FAR will decrease, but FRR will increase; this results in a more conservative system. On the other hand, moving t to the left will cause FRR to decrease, but FAR will increase, causing more impostors attempting to access the system. Based on Fig. 1, the decidability of the system accuracy is determined by how much overlap there is between the two distributions, i.e., genuine and impostor distributions, which represent the quantitative measurement of intraclass and interclass variations, respectively.

In this paper, we employ normalized dissimilarity measures $\gamma = 1 - \boldsymbol{x}^T\boldsymbol{y}$ in the analysis, where $\boldsymbol{x}$ and $\boldsymbol{y}$ are the face feature vector. Note that if the feature vector is properly normalized, the inner product $1 - \boldsymbol{x}^T\boldsymbol{y}$ that was used in the demonstration is directly related to similarity measures such as cosine angle and dissimilarity measures such as the Euclidean distance of the vectors; thus, their statistical properties are retained [22]. For instance, $\cos\theta = x^T y$, and $\varepsilon(x, y) = \sqrt{\sum_i (x_i - y_i)^2} = \sqrt{\sum_i x_i^2 + \sum_i y_i^2 - 2\sum_i x_i y_i} = \sqrt{2 - 2\boldsymbol{x}^T\boldsymbol{y}}$.

Next, we show that the dissimilarity measures are preserved after RP in genuine distribution, whereby the face features from an individual are compared. On the other hand, the interclass (impostor distribution) is amplified if different random subspaces are engaged (legitimate-token scenario). Then, we will combine all the observations together to infer the recognition performance of MRP in three different scenarios, i.e., the legitimate-token, stolen-token, and stolen-biometrics scenarios.

1) Genuine Distribution: We let $\boldsymbol{u} = \sqrt{(1/m)}\boldsymbol{R}\boldsymbol{x}$ and $\boldsymbol{v} = \sqrt{(1/m)}\boldsymbol{R}\boldsymbol{y}$, $\gamma_R = 1 - \boldsymbol{u}^T\boldsymbol{v}$, $\gamma = 1 - \boldsymbol{x}^T\boldsymbol{y}$, and $\delta = \gamma - \gamma_R$. Thus, we have

$$\delta = (\boldsymbol{u}^T\boldsymbol{v} - \boldsymbol{x}^T\boldsymbol{y}) = (1/m)\boldsymbol{x}^T\boldsymbol{R}^T\boldsymbol{R}\boldsymbol{y} - \boldsymbol{x}^T\boldsymbol{y}.$$

Then

$$\mathrm{E}[\delta] = (1/m)\boldsymbol{x}^T\mathrm{E}[\boldsymbol{R}^T\boldsymbol{R}]\boldsymbol{y} - \boldsymbol{x}^T\boldsymbol{y}.$$

From (3), we have

$$\mathrm{E}[\delta] = 0. \tag{8}$$

For $\mathrm{var}[\delta] = \mathrm{var}[\boldsymbol{u}^T\boldsymbol{v} - \boldsymbol{x}^T\boldsymbol{y}]$

$$\mathrm{var}[\delta] = \frac{1}{m}\left[\sum_{i=1}^{m} x_i^2 y_i^2 + \sum_{i=1}^{m} x_i^2 \sum_{j=1}^{m} y_j^2\right]. \tag{9}$$

If both $\boldsymbol{x}$ and $\boldsymbol{y}$ are normalized to unit length, thus, $\sum_i x_i^2 \sum_j y_j^2 = 1$, and $\sum_i x_i^2 y_i^2 < 1$; then, we have

$$\mathrm{var}[\delta] < 2/m. \tag{10}$$

The proof of (9) is given in the Appendix.

The preceding discussion shows that the error δ of the dissimilarity measures that were produced by the RP technique is zero, on average, and the variance is less than the inverse of m, which is the dimension of RP multiplied by 2 if the feature vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ are normalized. This indicates that the genuine distribution will remain to be the same as before and after RPs, particularly, when m is large. According to the JL lemma, the lower bound of m for this to happen is $m = (4\ln k)/(\varepsilon^2/2 - \varepsilon^3/3)$. However, in our experiments, we find that, when $m \approx n$, where n is the dimension of the feature vector, the preservation effect is attained. This analysis applies to genuine distributions in the legitimate-token, stolen-token, and stolen-biometrics scenarios.

2) Impostor Distribution: To the other spectrum, we consider the impostor distribution behavior in the legitimate-token and stolen-biometrics scenarios, where difference subspaces $\boldsymbol{R}$ and $\boldsymbol{S}$ are used. Note that impostor distribution for the stolen-token scenario would be identical to the genuine distribution discussed previously.

For instance, let $\boldsymbol{u} = \sqrt{(1/m)}\boldsymbol{R}\boldsymbol{x}$ and $\boldsymbol{v} = \sqrt{(1/m)}\boldsymbol{S}\boldsymbol{y}$; then, $\gamma_R = 1 - \boldsymbol{u}^T\boldsymbol{v}$. Thus, the impostor distribution mean is given by

$$\begin{aligned}\mu_i &= \mathrm{E}[\gamma_i] = \mathrm{E}[1 - \boldsymbol{u}^T\boldsymbol{v}]\\ &= \mathrm{E}\left[1 - (1/m)\boldsymbol{x}^T\boldsymbol{R}^T\boldsymbol{S}\boldsymbol{y}\right]\\ &= 1 - (1/m)\boldsymbol{x}^T\boldsymbol{y}\mathrm{E}[\boldsymbol{R}^T\boldsymbol{S}].\end{aligned}$$

From (6), we obtain

$$\mu_i = 1 \tag{11}$$

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and the impostor distribution variance $\sigma_i^2 = \mathrm{var}[\gamma_i] = \mathrm{var}[1 - \boldsymbol{u}^T\boldsymbol{v}] = \mathrm{var}[\boldsymbol{u}^T\boldsymbol{v}]$; we then have

$$\sigma_i^2 = \frac{1}{m}\left[\sum_{i=1}^{m} x_i^2 \sum_{j=1}^{m} y_j^2\right]. \tag{12}$$

If both $\boldsymbol{x}$ and $\boldsymbol{y}$ are normalized to unit length, $\sum_{i=1}^{m} x_i^2 \sum_{j=1}^{m} y_j^2 = 1$; hence

$$\sigma_i^2 \leq 1/m \text{ or standard deviation } \sigma_i \leq \sqrt{1/m}. \tag{13}$$

The proof of (12) is given in the Appendix.

In the legitimate-token scenario, we anticipate that the genuine curve will remain the same, the impostor curve will be centered at 1, and its profile envelope will shrink when m grows large, as depicted in Fig. 1. Therefore, the genuine and impostor distributions become separated, as m increases. The clear separation of the genuine-impostor distribution indicates zero FAR and FRR, and hence leads to zero error rate in the legitimate-token scenario.

Conversely, when the token is stolen, it means that different $\boldsymbol{x}$s are projected onto random subspace $\boldsymbol{R}$. Note that this is similar to the genuine analysis in Section III-B1, except that the feature vectors are from different users. It is obvious that the impostor (*pseudogenuine Type-I distribution* hereinafter) distribution will revert to its original state, as the dissimilarity measures are preserved, as indicated in Fig. 1. This is true when m is large.

On the other hand, for the stolen-biometrics scenario, the net effect is essentially equivalent to the projection of the compromised biometrics features onto multiple random subspaces; hence, it is expected that the impostor distribution (*pseudogenuine Type-II distribution* hereinafter) will peak at 1 and its standard deviation follows $1/\sqrt{m}$. We will verify all the assertions in the next section.

IV. Experiment and Discussion

A. Experiment Setup

In the experiments, EigenFace [24] is adopted as the face feature extractor, and $\gamma = 1 - \boldsymbol{x}^T\boldsymbol{y}$ is used as the dissimilarity measure, where $\boldsymbol{x}$ and $\boldsymbol{y}$ are normalized. For the eigenbasis training, we use 400 images (40 subjects with ten images each) from the ORL Face Database (http://www.uk.research.att.com/face data base.html). We evaluate the assertions by using the publicly available Facial Recognition Technology Face Database [25]. For the experiment, we randomly selected a subset of 600 users, each having four essentially frontal images with variations in pose (i.e., within ± 15 angle rotations out of plane), scale, and illumination. Many of these images were taken over an extended period and are highly varied, i.e., in terms of eyewear (absence and presence thereof) and illumination.

As the focus of this paper is on the efficacy with respect to genuine versus impostor classification, it is important to incorporate preprocessing mechanisms that contribute to recognition robustness. To this end, we performed geometrical normalization in order to establish correspondence between face images to be compared. The procedure is based on automatic location of the eye positions, from which various parameters (i.e., rotation, scaling, and translation) are used to extract the central part of the face from the original data set image. For the following experiments, all faces were subjected to DynaFace geometric normalization [26].

To generate the impostor distribution, the first MRP of each subject is matched against the first MRP of all other subjects, and the same matching process was repeated for subsequent MRPs, leading to $(600 \times 599)/2 \times 4 = 718\,800$ impostor attempts. For the genuine distribution, each MRP of each subject is matched against all other MRPs of the same subject, leading to 3600 $((3 \times 4)/2$ attempts of each subject $\times\ 600)$. For the pseudogenuine I distribution, we take the worst case, where the impostors always manage to steal the genuine token. In other words, only a set of PRN is mixed with all the facial images, and the matching is performed according to the impostor match described previously. Pseudogenuine II distribution is generated by taking other biometrics features from the data set mixed with the corresponding impostor PRN (impostor's own token). The genuine versus impostor distributions and the genuine versus pseudogenuine I and II distributions enable us to calculate the equal error rate (EER) for the genuine-token, stolen-token, and stolen-biometrics scenarios. EER refers to the average value of two error rates, i.e., FAR and FRR. We repeat the same process 20 times, and the results are averaged to reduce the statistical frustration that is caused by the different sets of random numbers. In this paper, pca and pcamrp-m denote EigenFace and MRP, respectively, with feature length m. Note that the feature length of pca n is 110 for this experiment. The experimental data are acquired for MRP with lengths of 20, 40, 60, 80, and 100. We also repeat the experiments with a better feature extractor such as SpectroFace, which combined the wavelet transform and Fourier transform [27] (spf and spfmrp-m hereinafter for spectroface and its MRP).

B. Performance Evaluation

1) EigenFace as Feature Extractor: From Fig. 2(a) and (b), the genuine distribution's mean and the standard deviation of pcamrp-m are identical to pca at $m = 100$. This suggests that the genuine distributions of MRP are preserved when $m \approx n$. On the contrary, the large deviation in the genuine's means and standard deviations for small m, such as $m = 20$, indicates the distortion, as discussed in Section III.

On the other hand, the experimental values of the impostor's and pseudogenuine II's means are packed around a theoretical mean of 1, regardless of m [Fig. 2(c)]. The empirical values of the impostor's standard deviation for various ms are also tightly tagged to the theoretical value $1/\sqrt{m}$, closely followed by the pseudogenuine II's standard deviation, as indicated in Fig. 2(d). The observation leads to the conclusion that the impostor and the pseudogenuine II distribution of MRP are independent of the feature extractor, and it solely relies on the MRP bit length m. This implies that the amplification of the interclass

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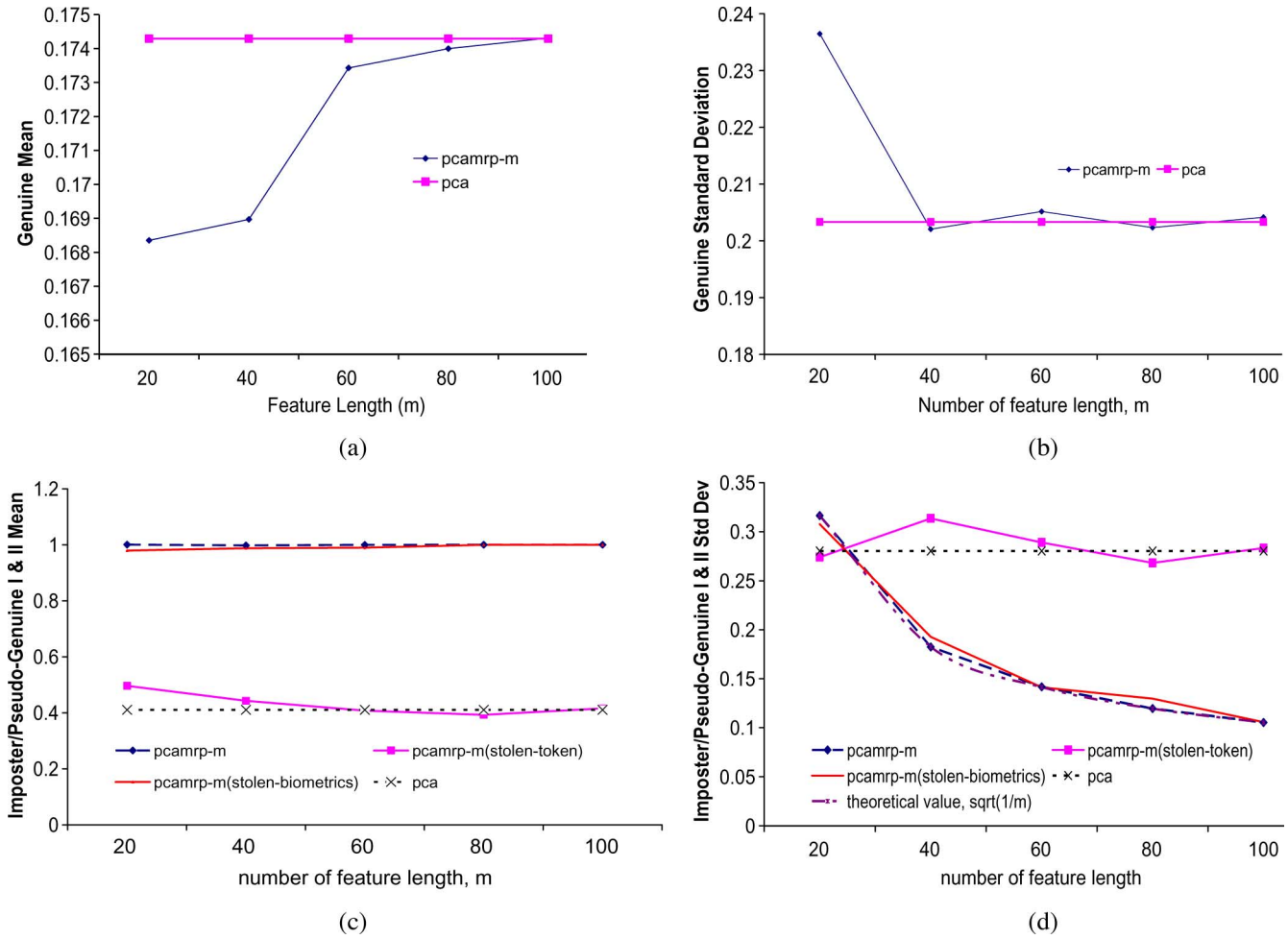

Fig. 2. (a) Mean and (b) standard deviation variations of the genuine distribution and (c) mean and (d) standard deviation variations of the impostor distribution according to MRP feature length m.

variation (the impostor/pseudogenuine II distribution) of MRP is controllable as long as $m < n$.

For the stolen-token scenario, we see that the pseudogenuine I distribution's mean is very close to pca, and the $1/\sqrt{m}$ rule is destroyed in standard deviation, as depicted in Fig. 2(c) and (d). This vindicated the inference that was described in Section III-B—the performance remains to be the same in the feature-vector level (EER $= 31.23\%$), which is one of the important requirements in cancelable biometrics formulation.

Fig. 3 illustrates the genuine versus impostor and genuine versus pseudogenuine I distributions of pca and their MRP counterparts, respectively. We observe that the genuine and impostor distributions of pca are highly similar to their corresponding MRP's genuine and pseudogenuine I distributions, respectively, with a slightly small m compare to n ($m = 100$ and $n = 110$ in the experiments).

2) Spectroface as Feature Extractor: Note that the performance of MRP is directly proportional to the quality of the feature extractor. Feature extractor quality will solely determine the performance of MRP in the stolen-token scenario. Anyhow, it would not be poorer than its original feature extractor. For the experiments with spf, we use a wavelet base with Daubechies filter order 7; level 1 is selected for spf and spfmrp-m. Note that the feature length for spf is $n = 127$, but

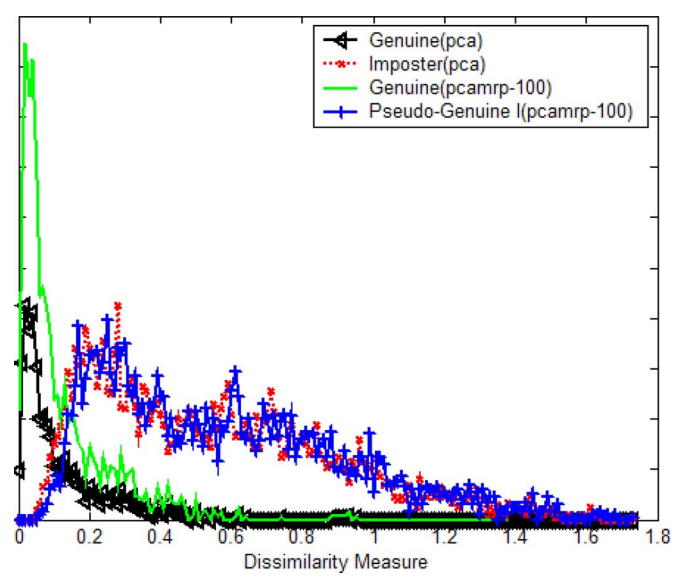

Fig. 3. Genuines and impostors for pca and pseudogenuine I distributions and pcamrp-100.

we set spfmrp-m to $m = 100$ for the fair comparisons with pcamrp-m. From Fig. 4, we notice that spf (EER $= 18.10\%$) is superior to pca (EER $= 31.23\%$) and the same as their MRP

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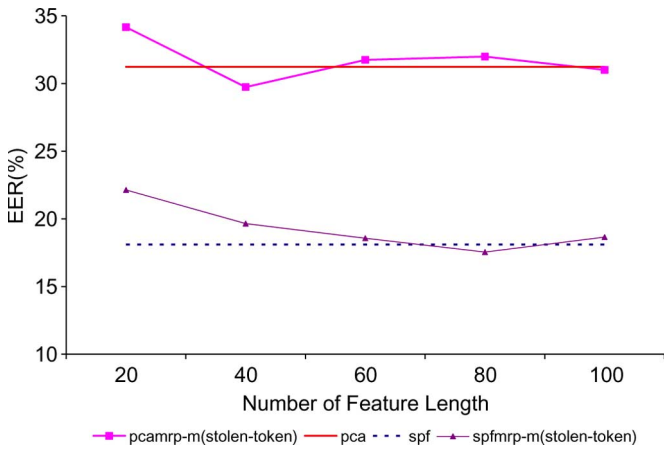

Fig. 4. Performance comparisons for pca and spf as well as their MRP counterparts pcamrp-m and spfmrp-m in the genuine-token, stolen-token, and stolen-biometrics scenarios.

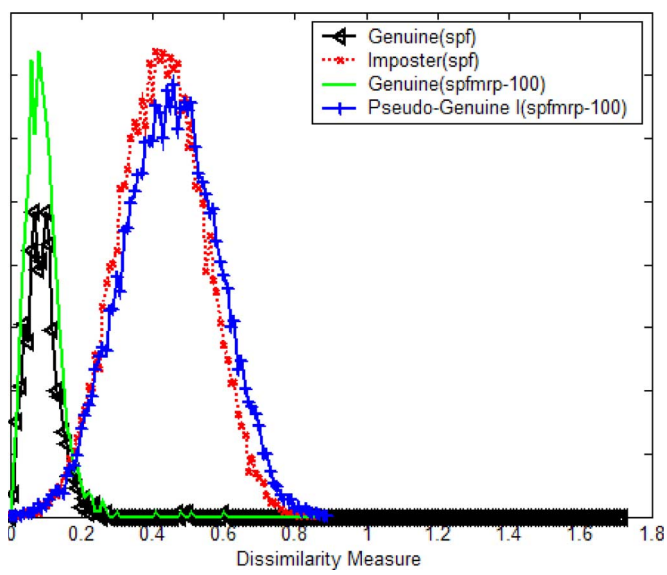

Fig. 5. Genuines and impostors for spf and pseudogenuine I distributions for spfmrp-100.

counterparts (EER $= 18.65\%$ for spfb-100 and EER $= 31.07\%$ for pcab-100) in the stolen-token scenario. Again, Fig. 5 depicts the genuine versus impostor and genuine versus pseudogenuine I distributions of spf and their MRP counterparts, respectively. We observe that the genuine and impostor distributions of spf are less overlapped compared those to pca in Fig. 3 but highly similar to their corresponding MRP's genuine and pseudogenuine I distributions, respectively. This recites our assertion that the MRP preserves the statistical properties of the dissimilarity measures of the biometrics feature extractor.

3) Euclidean Distance as Dissimilarity Measure: From the analysis in Section III, the statistical preserving transformation is essentially ruled by the choice of normalized dissimilarity measure, i.e., $\gamma_g = 1 - \mathbf{x}^T \boldsymbol{y}$. Despite that, we also inferred that the other similarity measures such as cosine angle and dissimilarity measure, and the Euclidean distance also perform the same. In this section, we use the Euclidean distance to preserve

TABLE I
PERFORMANCE COMPARISONS WITH PCA, SPF, AND THEIR MRP STOLEN-TOKEN COUNTERPARTS AT $m = 100$, WITH DISSIMILARITY MEASURES $\gamma_g = 1 - \mathbf{x}^T \boldsymbol{y}$ AND NORMALIZED EUCLIDEAN DISTANCE

Methods	EER(%)	
	Normalized Euclidean Distance	$\gamma_g = 1 - x^T y$
pca	25.51	31.24
pcamrp(stolen-token)	25.77	31.01
spf	16.11	18.10
spfmrp (stolen-token)	15.97	18.65

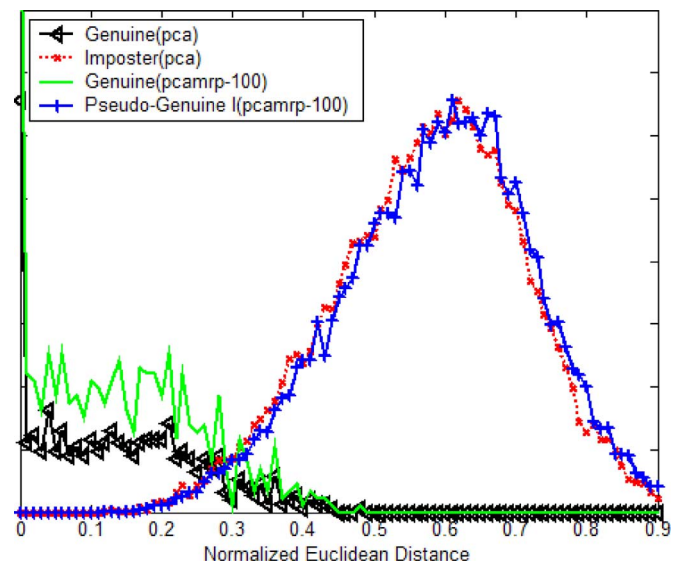

Fig. 6. Genuines and impostors for pca and pseudogenuine I distributions for pcamrp-100 with normalized Euclidean distance.

the statistical properties of the dissimilarity measures of the biometrics feature extractor. We repeat the same experiments, as discussed previously, but we use the normalized Euclidean distance instead. From Table I, we observe that pca, spf, and their MRP stolen-token counterpart performed equally well at about EER $\approx 25\%$ and EER $\approx 16\%$. This implies that the normalized Euclidean distance is another dissimilarity measure that can be used in MRP, but it has the advantage compared with γ_g for its better discrimination capacity. Figs. 6 and 7 illustrate their corresponding genuine-impostor distributions.

However, the MRP can be generalized for other distance-based classifiers such as k-NN as well as more powerful classifiers, such as support vector machines [22] and probabilistic neural networks (PNN). This is because the original data points are completely quantified by the inner product in the Lagrange dual optimization problem, i.e.,

$$\min_\alpha 1/2 \Sigma_i \Sigma_j y_i y_j \alpha_i \alpha_j x_i x_j - \Sigma_i \alpha_i$$

$$\text{s.t.} \quad \Sigma_i y_i \alpha_i = 0 \text{ and } \alpha_i \geq 0$$

where $y_i = \pm 1$ is the class label of point i.

Similarly, for PNN, the kernel φ_j is also described by using $\varphi_j = \exp(-\Sigma i(x_i^T y_{ij}))$.

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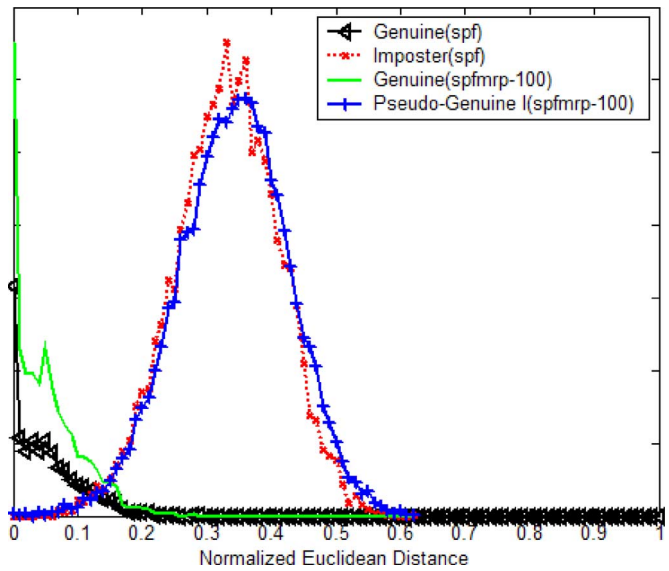

Fig. 7. Genuines and impostors for spf and pseudogenuine I distributions for spfmrp-100 with normalized Euclidean distance.

C. Discussion

From the preceding sections, we observed that MRP fulfills the requirement of cancelable biometrics in terms of performance, even in the stolen-token scenario, whereby the performance is retained as that in the feature-vector level. In practice usage, we should set the system threshold t, which is used to decide the acceptance/rejection of the users according to the feature-level performance (or the stolen-token performance profile), instead of others scenarios. Nevertheless, recall that our results need to be an important contribution to help preserve the privacy of the biometrics data and enable the enrolled template to be replaced in the event of template compromise.

In one implementation scenario, an MRP feature can be stored on an offline token or centered database during enrollment, and then used as a reference for user authentication. If the token is stolen or lost, it can be simply reenrolled using another PRN sequence to produce a refreshed template. The adopted PRN sequence is generated using a seed from a tamper-proof physical device (USB token or smartcard), which can be its serial number. Note that the PRN sequence is unique if they are generated from different seeds. During the verification stage, the biometrics feature is mixed with the legitimate PRN, and the resulting MRP template is compared with the enrolled template. However, more secure two-factor MRP deployment can be done by using challenge-response and secret-sharing techniques, or some cryptographic protocols such as the extension of the Diffie–Hellman protocol, Schnorr signature, and Zheng signcryption protocols, as reported in [16].

The MRP is indeed not restricted to the user-specific PRN but can also apply in common PRN setting. In this formulation, we use one set of PRN to conceal all biometrics features to eliminate the stolen-token concern and to readily apply it in the identification configuration. However, it would be susceptible to an attack, which enables recovery of the biometrics feature through multiple extractions of the MRP template.

One may argue that the PRN overpowers the biometrics in the MRP formulation, and the resulting biometrics role is nullified. However, we content both components (PRN + biometrics) play equally important roles in the MRP. For instance, if the PRN overtakes biometrics, the most apparent effect is the zero-mean and standard deviation occurrences in the genuine distribution. Furthermore, without the presence of PRN, sole biometrics suffers from nonrevocable and privacy invasion issues, which are the primary concerns of the cancelable biometrics, while sole token usage is susceptible to repudiation.

V. Cancelable and Noninvertible Property of MRP

We also examine the cancelable property of MRP by evaluating whether MRP A and MRP B (with same pca feature) are correlated. In other words, we wish to avoid the old MRP from falling into the region of acceptance of the refreshed MRP. In this case, we let $\boldsymbol{u} = \sqrt{(1/m)}\boldsymbol{R}\boldsymbol{x}$, $\boldsymbol{v} = \sqrt{(1/m)}\boldsymbol{S}\boldsymbol{x}$, and $\boldsymbol{R} \neq \boldsymbol{S}$; then, the correlation of $\boldsymbol{u}$ and $\boldsymbol{v}$ is

$$\boldsymbol{u}^T\boldsymbol{v} = 1/m\ \boldsymbol{x}^T\boldsymbol{R}^T\boldsymbol{S}\boldsymbol{x}.$$

According to (8)

$$\mathrm{E}[\boldsymbol{u}^T\boldsymbol{v}] = 0$$

and

$$\mathrm{var}[\boldsymbol{u}^T\boldsymbol{v}] = 1/m.$$

This implies that there is no correlation for the old MRP and the refreshed MRP when m is large. Experimentally, the evaluation is exactly the same as the impostor distribution of the stolen-biometrics scenario, where different PRNs are mixed with the same pca feature. As shown in Fig. 2(c) and (d), the performance is very close to the genuine-token scenario, where the impostor distribution is centered at 0.5. This implies that the refreshed MRP is not resembling the old MRP, particularly, when $m \approx n$; hence, PRN refreshment is equivalent to issues of the user in a new template.

The security analysis of the noninvertible property can be carried out by looking at (1): $\boldsymbol{v} = \boldsymbol{R}\boldsymbol{x}$, where $\boldsymbol{R}$ is an $m \times n$ orthonormal random matrix and $m \leq n$. $\boldsymbol{v}$ can be regarded as a set of underdetermined systems of linear equations (more unknowns than equations). Therefore, it is impossible to find the *exact values of all the elements* in $\boldsymbol{x}$ by solving an underdetermined linear equation system in $\boldsymbol{v} = \boldsymbol{R}\boldsymbol{x}$ if $m < n$, based on the premise that the possible solutions are infinite [28].

VI. Concluding Remark

A generic cancelable biometrics formulation that is coined as MRPs is proposed in this paper. MRP is carried out by first transforming raw biometrics data into a fixed-length feature vector, followed by the noninvertible random subspace projection. The random subspace is constituted by the independent, zero-mean, and unit-variance Gaussian-distributed random bases, which can be generated from the user-specific

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seeded pseudorandom numbers generator that is stored in a token or centered database. MRP can be revoked through the pseudorandom numbers replacement, so that a new template can be generated instantly, and its noninvertible property helps to preserve the privacy of the user biometrics data.

MRP offers significant performance improvement in the verification setting, such as the separation of the genuine and impostor distributions for the legitimate-token scenario. This is accomplished through the multiple-RP effect of preserving intraclass variations while amplifying interclass variations via projection onto uncorrelated random subspaces. There is an important proviso, i.e., that the verification performance also depends on the quality of the feature extractor. We also demonstrated that the MRP could function well without compromising the recognition performance in the event of compromised token or biometrics.

APPENDIX

Proof of (9): Let $\boldsymbol{R}$ be an $m \times n\,(m < n)$ random matrix such that each entry $r_{ij} \in \boldsymbol{R}$ is i.i.d. drawn from $N(0,1)$, and let ε_{ij} be the entry of $\boldsymbol{R}^T\boldsymbol{R}$. $\boldsymbol{u}$ and $\boldsymbol{v}$ are the projected face feature vectors $\boldsymbol{x}$ and $\boldsymbol{y}$, respectively, in $\Re^m$.

Furthermore, let $\boldsymbol{u} = 1/(\sqrt{m})\boldsymbol{R}\boldsymbol{x}$ and $\boldsymbol{v} = 1/(\sqrt{m})\boldsymbol{R}\boldsymbol{y}$.

Then, the variance distortion of $\boldsymbol{u}^T\boldsymbol{v} - \boldsymbol{x}^T\boldsymbol{y}$ is

$$\begin{aligned}\mathrm{var}[\delta] &= \mathrm{var}[\boldsymbol{u}^T\boldsymbol{v} - \boldsymbol{x}^T\boldsymbol{y}] \\ &= \frac{1}{m}\left[\sum_{i=1}^{m} x_i^2 y_i^2 + \sum_{i=1}^{m} x_i^2 \sum_{j=1}^{m} y_j^2\right].\end{aligned}$$

Proof: To compute the variance of distortion δ, $\boldsymbol{u}^T\boldsymbol{v}$ can be decomposed into diagonal and off-diagonal entries of ε_{kj}, such as

$$\begin{aligned}\boldsymbol{u}^T\boldsymbol{v} &= (1/m)\boldsymbol{x}^T\boldsymbol{R}^T\boldsymbol{R}\boldsymbol{y} \\ &= \frac{1}{m}\left[\sum_i x_i\varepsilon_{ii}y_i + \sum_{i\neq j} x_i\varepsilon_{ij}y_j\right].\end{aligned}$$

Let $\alpha = \sum_i x_i\varepsilon_{ii}y_i$ and $\beta = \sum_{i\neq j} x_i\varepsilon_{ij}y_j$. Then

$$\mathrm{var}[\boldsymbol{u}^T\boldsymbol{v}] = \mathrm{var}[\alpha] + \mathrm{var}[\beta] + \mathrm{cov}[\alpha,\beta].$$

Now, we consider

$$\mathrm{cov}[\alpha,\beta] = \mathrm{E}[\alpha\beta] - \mathrm{E}[\alpha]\mathrm{E}[\beta].$$

Since $\mathrm{E}[\varepsilon_{ij}] = 0\ \forall i, j, i \neq j$, $\mathrm{E}[\beta] = 0$. Hence

$$\begin{aligned}\mathrm{cov}[\alpha,\beta] &= \mathrm{E}[\alpha\beta] - 0 \\ &= \frac{1}{m^2}\mathrm{E}\left[\sum_{i=1} x_i y_i \varepsilon_{ii} \mathrm{x} \sum_{r\neq s} x_r y_s \varepsilon_{rs}\right].\end{aligned}$$

It is straightforward to verify that $\mathrm{E}[\varepsilon_{ii}\varepsilon_{rs}] = 0$ when $r \neq s$. Therefore, $\mathrm{cov}[\alpha,\beta] = 0$.

Then

$$\mathrm{var}[\boldsymbol{u}^T\boldsymbol{v}] = \mathrm{var}[\alpha] + \mathrm{var}[\beta].$$

We redefine $\alpha = \sum_{i=1}^{m} x_i y_i \varepsilon_{ii}$ and decompose β further into upper and lower triangle off-diagonal entries of ε_{kj}, i.e.,

$$\beta = \sum_{i=1}^{m-1}\sum_{j=i+1}^{m} x_i y_j \varepsilon_{ij} + \sum_{i=2}^{m}\sum_{j=1}^{i-1} x_i y_j \varepsilon_{ij}.$$

We can easily verify that

$$\mathrm{var}[\beta] = \mathrm{var}\left[\sum_{i=1}^{m-1}\sum_{j=i+1}^{m} x_i y_j \varepsilon_{ij}\right] + \mathrm{var}\left[\sum_{i=2}^{m}\sum_{j=1}^{i-1} x_i y_j \varepsilon_{ij}\right]$$

where

$$\mathrm{cov}\left[\sum_{i=1}^{m-1}\sum_{j=i+1}^{m} x_i y_j \varepsilon_{ij}, \sum_{i=2}^{m}\sum_{j=1}^{i-1} x_i y_j \varepsilon_{ij}\right] = 0.$$

Then

$$\begin{aligned}\mathrm{var}[\boldsymbol{u}^T\boldsymbol{v}] &= \frac{1}{m^2}\left[\mathrm{var}\left[\sum_{i=1}^{m} x_i y_i \varepsilon_{ii}\right] + \mathrm{var}\left[\sum_{i=1}^{m-1}\sum_{j=i+1}^{m} x_i y_j \varepsilon_{ij}\right]\right. \\ &\qquad \left. + \mathrm{var}\left[\sum_{i=2}^{m}\sum_{j=1}^{i-1} x_i y_j \varepsilon_{ij}\right]\right] \\ &= \frac{1}{m^2}\left[\sum_{i=1}^{m}(x_i y_i)^2 \mathrm{var}[\varepsilon_{ii}] + \sum_{i=1}^{m-1}\sum_{j=i+1}^{m} x_i^2 y_j^2 \mathrm{var}[\varepsilon_{ij}]\right. \\ &\qquad \left. + \sum_{i=2}^{m}\sum_{j=1}^{i-1} x_i^2 y_j^2 \mathrm{var}[\varepsilon_{ij}]\right].\end{aligned}$$

From (5), we have

$$\mathrm{var}(\varepsilon_{ij}) = \begin{cases} 2m, & \text{if } i = j \\ m, & \text{if } i \neq j. \end{cases}$$

Then

$$\begin{aligned}\mathrm{var}[\boldsymbol{u}^T\boldsymbol{v}] &= \frac{1}{m}\left[2\sum_{i=1}^{m}(x_i y_i)^2 + \sum_{i=1}^{m-1}\sum_{j=i+1}^{m} x_i^2 y_j^2 + \sum_{i=2}^{m}\sum_{j=1}^{i-1} x_i^2 y_j^2\right] \\ &= \frac{1}{m}\left[\sum_{i=1}^{m} x_i^2 y_i^2 + \left(\sum_{i=1}^{m} x_i^2 y_i^2 + \sum_{i=1}^{m-1}\sum_{j=i+1}^{m} x_i^2 y_j^2\right.\right. \\ &\qquad \left.\left. + \sum_{i=2}^{m}\sum_{j=1}^{i-1} x_i^2 y_j^2\right)\right] \\ &= \frac{1}{m}\left[\sum_{i=1}^{m} x_i^2 y_i^2 + \sum_{i=1}^{m}\sum_{j=1}^{m} x_i^2 y_j^2\right].\end{aligned}$$

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Therefore

$$\mathrm{var}\left[\boldsymbol{u}^T\boldsymbol{v}-\boldsymbol{x}^T\boldsymbol{y}\right]=\mathrm{var}\left[\boldsymbol{u}^T\boldsymbol{v}\right]$$
$$=\frac{1}{m}\left[\sum_{i=1}^{m}x_i^2y_i^2+\sum_{i=1}^{m}x_i^2\sum_{j=1}^{m}y_j^2\right].$$

Proof of (12): Let $\boldsymbol{R}$ and S be two $m\times n\,(m<n)$ uncorrelated random matrices such that each entry $r_{ij}\in\boldsymbol{R}$ and $s_{ij}\in\boldsymbol{S}$ are i.i.d. drawn from $N(0,1)$ and η_{ij} be the entry of $\boldsymbol{R}^T\boldsymbol{S}$. $\boldsymbol{u}$ and $\boldsymbol{v}$ are the projected face feature vectors $\boldsymbol{x}$ and $\boldsymbol{y}$, respectively, in $\Re^m$. It can be proven that η_{ij} is approximately Gaussian, $\mathrm{E}[\eta_{ij}]=0$, and $\mathrm{var}[\eta_{ij}]=m\ \forall i,\,j$.

Therefore, the impostor distribution variance σ_i^2 is

$$\sigma_i^2=\mathrm{var}[\gamma_i]=\mathrm{var}[1-\boldsymbol{u}^T\boldsymbol{v}]$$
$$=\mathrm{var}[\boldsymbol{u}^T\boldsymbol{v}]$$
$$=\frac{1}{m}\left[\sum_{i=1}^{m}x_i^2\sum_{j=1}^{m}y_j^2\right].$$

Proof: Similarly, we decompose $\boldsymbol{u}^T\boldsymbol{v}$ into the terms of diagonal, upper, and lower triangle off-diagonal entries of η_{ij}.

From

$$\boldsymbol{u}^T\boldsymbol{v}=(1/m)\boldsymbol{x}^T\boldsymbol{R}^T\boldsymbol{S}\boldsymbol{y}$$
$$=\frac{1}{m}\left[\sum_{i=1}^{m}x_iy_i\eta_{ii}+\sum_{i=1}^{m-1}\sum_{j=i+1}^{m}x_iy_j\eta_{ij}+\sum_{i=2}^{m}\sum_{j=1}^{i+1}x_iy_j\eta_{ij}\right].$$

Since $\mathrm{E}[\eta_{ij}]=0\ \forall i,\,j$, follow the similar steps as before; we have

$$\mathrm{var}[\boldsymbol{u}^T\boldsymbol{v}]=\frac{1}{m^2}\left[\mathrm{var}\left[\sum_{i=1}^{m}x_iy_i\eta_{ii}\right]\right.$$
$$\left.+\ \mathrm{var}\left[\sum_{i=1}^{m-1}\sum_{j=i+1}^{m}x_iy_j\eta_{ij}\right]+\ \mathrm{var}\left[\sum_{i=2}^{m}\sum_{j=1}^{i-1}x_iy_j\eta_{ij}\right]\right].$$

From (5), we have $\mathrm{var}[\eta_{ij}]=m\ \forall i,\,j$; then

$$\mathrm{var}[\boldsymbol{u}^T\boldsymbol{v}]=\frac{1}{m}\left[\sum_{i=1}^{m}x_i^2y_i^2+\sum_{i=1}^{m-1}\sum_{j=i+1}^{m}x_i^2y_j^2+\sum_{i=2}^{m}\sum_{j=1}^{i-1}x_i^2y_j^2\right]$$
$$=\frac{1}{m}\left[\sum_{i=1}^{m}\sum_{j=1}^{m}x_i^2y_j^2\right].$$

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